

1. RUN CONFIGURATION

|                  |          |                |                |                |
|------------------|----------|----------------|----------------|----------------|
| Frames Processed | Duration | Processing FPS | Avg Processing | Max Processing |
| 450              | 1.96 s   | 229.0          | 2.13 ms        | 5.62 ms        |

2. TRACKING PERFORMANCE

|                          |                                  |                              |
|--------------------------|----------------------------------|------------------------------|
| Acquisition Time         | Average Tracking Error           | Maximum Tracking Error       |
| 0.03 s                   | 2.68 px                          | 6.08 px                      |
| <small>&lt;= 2 s</small> |                                  | <small>&lt;= 10 px</small>   |
| RMSE Tracking Error      | Centroid Error Avg/Max           | Lock Retention               |
| 2.97 px                  | 2.68 / 6.08 px                   | 100.00 %                     |
|                          | <small>ground truth only</small> |                              |
| Target Loss              | Re-acquisition Avg/Max           | Successful Re-acq.           |
| 0.00 %                   | —                                | 0                            |
| <small>&lt; 5 %</small>  | <small>&lt;= 1 s</small>         | <small>0 loss events</small> |

3. PS REQUIREMENT CHECK

| Parameter            | Measured  | Requirement | Status |
|----------------------|-----------|-------------|--------|
| Acquisition Time     | 0.03      | <= 2.0 s    | PASS   |
| Tracking Error (Max) | 6.08      | <= 10 px    | PASS   |
| Center Error (Final) | 9.22      | <= 10 px    | PASS   |
| Target Loss          | 0.00 %    | < 5%        | PASS   |
| Re-acquisition Time  | 0.13      | <= 1.0 s    | PASS   |
| Processing Speed     | 229.0 FPS | >= 20 FPS   | PASS   |

Ground truth was available: centroid accuracy is measured.

OVERALL: PASS

Source: benchmark video: synthetic\_fsoc\_beacon.mp4 | Frames: 450 | FPS: 229.0